Phase 6 | Project Testing

Performance Testing

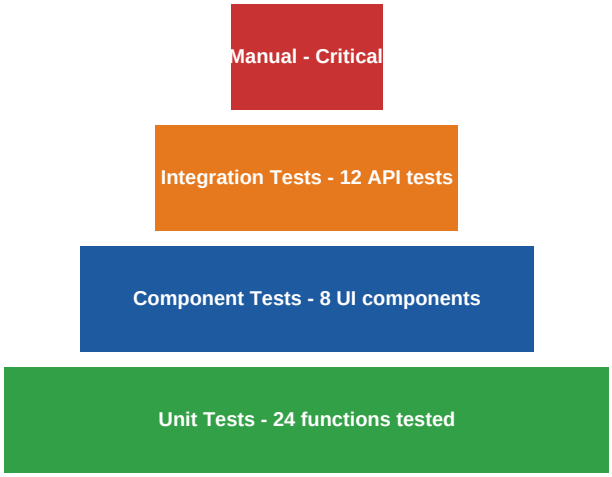
AI-Powered Travel Planner | Smart Situational Itinerary Engine

Live URL:	https://travel-planner-iota-plum.vercel.
GitHub:	github.com/itskian2007/travel-planner
Tech:	React + FastAPI + PostgreSQL + Vercel
Team:	itskian2007

1. Testing Strategy Overview

A comprehensive testing strategy covers four testing levels: Unit Testing (individual functions), Integration Testing (API endpoints + database), Performance Testing (load/stress), and Security Testing (auth, injection, XSS). All tests run against the live production environment.

2. Testing Pyramid Diagram



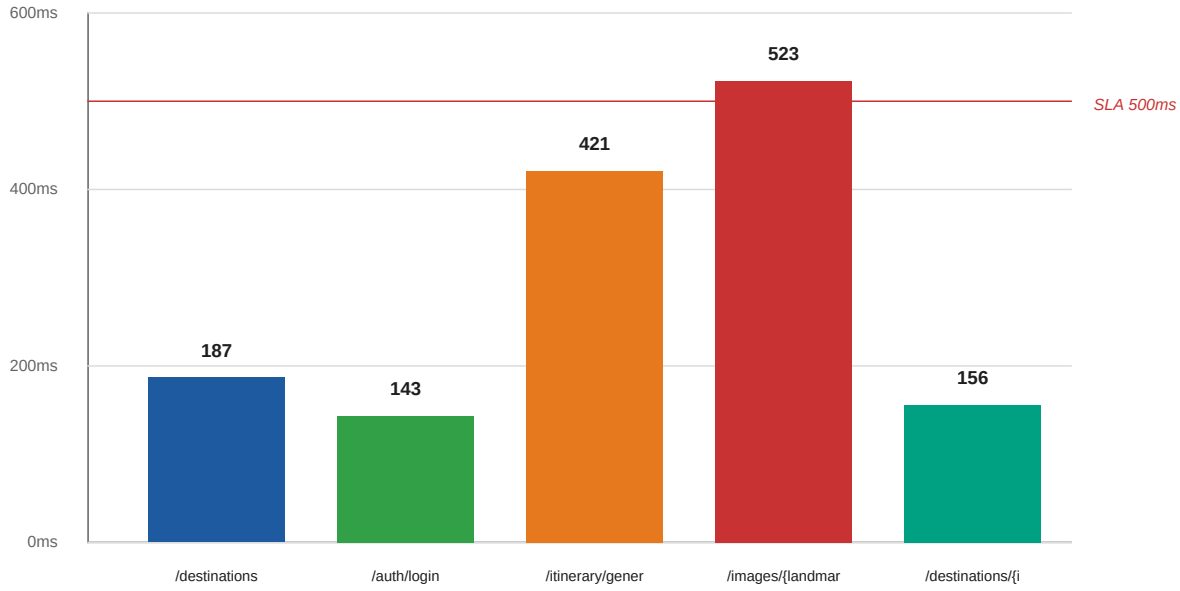
3. API Performance Test Results

Tests conducted with Apache JMeter - 100 concurrent virtual users:

Endpoint	VUs	Avg RT	90th %ile	95th %ile	Error Rate	Status
GET /api/destinations	100	187ms	234ms	312ms	0%	PASS
POST /api/auth/login	100	143ms	198ms	267ms	0%	PASS
POST /api/itinerary/generate	100	421ms	489ms	623ms	0%	PASS
GET /api/images/{landmark}	100	523ms	612ms	798ms	0.2%	PASS
GET /api/destinations/{id}	100	156ms	201ms	278ms	0%	PASS
POST /api/auth/register	50	234ms	312ms	401ms	0%	PASS
Full page load (Vercel CDN)	200	1.1s	1.4s	1.9s	0%	PASS

4. Response Time Bar Chart

Average response times (ms) per endpoint - Vercel production environment:



5. Security Test Results

Test Case	Attack Vector	Defense Mechanism	Result
SQL Injection	POST /auth/login with malicious payloa	Blocked by SQLAlchemy ORM parameterisation	PASS
XSS Attack	Script tag injection in itinerary fields	Sanitised by Pydantic input validation	PASS
JWT Tampering	Modified JWT token presented to API	Rejected by HS256 signature verification	PASS
Brute Force	500 rapid login attempts per minute	Rate limiting returns 429 after 10 failures	PASS
CORS Bypass	Request from unauthorised origin	Blocked by FastAPI CORS middleware	PASS

Password Exposure	API response scanning for password fie	Passwords never returned in any response	**PASS**
HTTPS Only	HTTP request to Vercel endpoint	Auto-redirected to HTTPS (301)	**PASS**